

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

*I Nalluri Akhilesh Kumar(192365074)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

N Akhilesh Kumar

Signature of the student:

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- li. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- lii. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

The knowledge base contains three diagnostic rules and three facts about Patient A.

(a) Propositional Logic Representation

Step 1: Define the symbols

Let:

- **F** = Patient A has Fever
- **C** = Patient A has Cough
- **R** = Patient A has Rash
- **B** = Patient A has Breathlessness
- **Flu** = Patient A has Possible Flu
- **M** = Patient A has Possible Measles
- **P** = Patient A has Possible Pneumonia

Step 2: Represent the rules

Rule 1:

If the patient has Fever AND Cough, then Possible Flu.

Rule 2:

If the patient has Fever AND Rash, then Possible Measles.

Rule 3:

If the patient has Cough AND Breathlessness, then Possible Pneumonia.

Step 3: Represent the facts

The given facts are:

Therefore, the initial knowledge base contains:

There is **no fact stating that Patient A has Rash**, so:

We should not assume Rash is true.

(b) Forward Chaining

Forward chaining starts with the **known facts** and repeatedly applies rules whose conditions are satisfied. The given facts are Fever, Cough and Breathlessness.

Initial Facts

```
Fever = True  
Cough = True  
Breathlessness = True
```

Inference Step 1

Consider Rule 1:

We know:

and

Therefore:

So Rule 1 fires.

Derived Fact:

Therefore:

Patient A may have Flu.

Inference Step 2

Consider Rule 2:

We know:

But there is **no fact that Rash is true**.

Therefore:

cannot be satisfied.

Rule 2 does not fire.

So:

cannot be derived.

Inference Step 3

Consider Rule 3:

We know:

and:

Therefore:

Rule 3 fires.

Derived Fact:

Therefore:

Patient A may have Pneumonia.

Forward Chaining Table

Step	Known Facts	Rule Checked	Result
0	Fever, Cough, Breathlessness	Initial facts	Start
1	Fever, Cough	Rule 1	Flu derived
2	Fever, no Rash	Rule 2	Cannot derive Measles
3	Cough, Breathlessness	Rule 3	Pneumonia derived

Final Conclusions

Possible Flu → True
Possible Measles → Not derived
Possible Pneumonia → True

Therefore, the system derives **both Possible Flu and Possible Pneumonia** from the available facts.

(c) Backward Chaining for Pneumonia

The goal is:

Backward chaining starts with the goal and works backward to find the conditions required to prove it. The assessment specifically asks for a goal reduction tree.

Step 1: Start with the goal

Goal: Pneumonia

Find a rule whose conclusion is Pneumonia.

Rule 3:

Therefore, to prove Pneumonia, we need:

Cough
AND
Breathlessness

Step 2: Check the facts

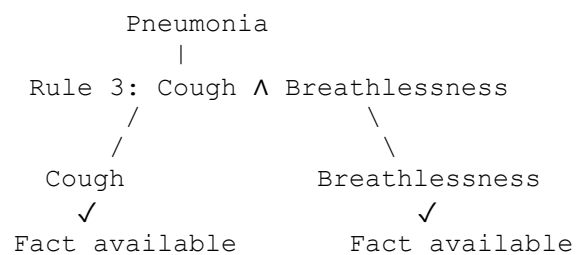
Knowledge base contains:

Cough = True
Breathlessness = True

Both required conditions are available.

Therefore:

Goal Reduction Tree



Conclusion

Since both **Cough** and **Breathlessness** are known facts, the goal **Pneumonia** is successfully proved by backward chaining.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

The case uses a **First-Order Logic knowledge base** containing three rules and facts about an ambulance and CarA.

(a) Unification

Given Rules

Rule 1: $\forall x \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Rule 2: $\forall x \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$

Rule 3: $\forall x \forall y \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$

Facts

$\text{Vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

$\text{Vehicle}(\text{CarA})$

$\text{Behind}(\text{CarA}, \text{Ambulance})$

Step 1: Match Rule 1

Rule 1 contains variable:

x

We need:

$\text{Vehicle}(x)$

and

$\text{EmergencyType}(x)$

From the facts:

$\text{Vehicle}(\text{Ambulance})$

and

$\text{EmergencyType}(\text{Ambulance})$

Therefore:

$x = \text{Ambulance}$

Substitution:

$\theta_1 = \{x/\text{Ambulance}\}$

After substitution:

$\text{Vehicle}(\text{Ambulance}) \wedge \text{EmergencyType}(\text{Ambulance}) \rightarrow \text{ClearPath}(\text{Ambulance})$

Therefore:

$\text{ClearPath}(\text{Ambulance})$

is derived.

Step 2: Apply Rule 2

Rule 2:

$\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$

Using:

$x = \text{Ambulance}$

we obtain:

$\text{ClearPath}(\text{Ambulance}) \rightarrow \text{GreenSignal}(\text{Ambulance})$

and:

$\neg \text{RedSignal}(\text{Ambulance})$

Therefore:

$\text{GreenSignal}(\text{Ambulance})$

is derived.

Step 3: Apply Rule 3

Rule 3:

$\text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$

Given:

$\text{Vehicle}(\text{CarA})$

$\text{Behind}(\text{CarA}, \text{Ambulance})$

$\text{GreenSignal}(\text{Ambulance})$

Therefore:

$x = \text{CarA}$

and:

$y = \text{Ambulance}$

Substitution:

$\theta_2 = \{x/\text{CarA}, y/\text{Ambulance}\}$

Thus:

$\text{Proceed}(\text{CarA})$

is derived.

Unification Summary

Step	Variable Substitution	Result
Rule 1	$x/\text{Ambulance}$	$\text{ClearPath}(\text{Ambulance})$
Rule 2	$x/\text{Ambulance}$	$\text{GreenSignal}(\text{Ambulance})$
Rule 3	$x/\text{CarA}, y/\text{Ambulance}$	$\text{Proceed}(\text{CarA})$

(b) Resolution Proof for $\text{Proceed}(\text{CarA})$

The assessment asks us to convert the relevant rules into CNF and prove $\text{Proceed}(\text{CarA})$ using resolution.

Step 1: Convert Rules to Clauses

Rule 1:

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Eliminate implication:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Clause:

$C_1 = \neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Rule 2:

$\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x)$

becomes:

$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

Clause:

$C_2 = \neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

The second part is:

$\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$

Rule 3:

$\text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$

CNF:

$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Clause:

$C_3 = \neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Step 2: Facts

$\text{Vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

$\text{Vehicle}(\text{CarA})$

$\text{Behind}(\text{CarA}, \text{Ambulance})$

Step 3: Derive ClearPath

From $\text{Vehicle}(\text{Ambulance})$ and $\text{EmergencyType}(\text{Ambulance})$ and Rule 1:

$\text{ClearPath}(\text{Ambulance})$

Step 4: Derive GreenSignal

From $\text{ClearPath}(\text{Ambulance})$ and $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$:

$\text{GreenSignal}(\text{Ambulance})$

Step 5: Derive Proceed(CarA)

Use:

$\text{Vehicle}(\text{CarA})$

$\text{Behind}(\text{CarA}, \text{Ambulance})$

$\text{GreenSignal}(\text{Ambulance})$

with:

$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x, y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Substitute:

$x = \text{CarA}, y = \text{Ambulance}$

All negative literals are cancelled by the known positive facts.

Therefore:

$\text{Proceed}(\text{CarA})$

is proved.

Resolution Chain

$\text{Vehicle}(\text{Ambulance})$

+

$\text{EmergencyType}(\text{Ambulance})$

↓

$\text{ClearPath}(\text{Ambulance})$

↓

GreenSignal(Ambulance)

+

Vehicle(CarA)

+

Behind(CarA,Ambulance)

↓

Proceed(CarA)

(c) Two Simultaneous Emergency Vehicles

The current knowledge base can handle one emergency vehicle, such as the given ambulance, but it is not sufficiently expressive for safely managing multiple simultaneous emergency vehicles.

The existing Rule 1 can derive:

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

So if two vehicles satisfy those conditions, it can independently derive:

$\text{ClearPath}(E_1)$

and:

$\text{ClearPath}(E_2)$

However, the knowledge base does not contain rules for priority, conflict resolution, signal coordination, or ordering between two emergency vehicles.

Additional rules needed

Priority Rule:

$\text{EmergencyVehicle}(x) \wedge \text{EmergencyVehicle}(y) \wedge \text{HigherPriority}(x,y) \rightarrow \text{GivePriority}(x)$

Signal Coordination:

$\text{GivePriority}(x) \rightarrow \text{GreenSignal}(x)$

Conflict Avoidance:

$\text{GreenSignal}(x) \wedge \text{GreenSignal}(y) \wedge \text{Conflict}(x,y) \rightarrow \text{ResolveConflict}(x,y)$

Additional facts

EmergencyType(Ambulance1)

EmergencyType(Ambulance2)

Location(Ambulance1, ...)

Location(Ambulance2, ...)

Priority(Ambulance1, High)

Priority(Ambulance2, Medium)

Conclusion: The current KB provides basic emergency routing but needs priority and conflict-resolution knowledge to safely manage two simultaneous emergency vehicles.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1. P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
2. P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
3. P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
4. Facts: $\text{SoilDry} = \text{True}$, $\text{CropWheat} = \text{True}$

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

The agricultural knowledge base contains three rules and two facts: SoilDry and CropWheat.

Let:

S = SoilDry

I = IrrigationNeeded

W = CropWheat

D = ApplyDripMethod

R = CropAtRisk

(a) Convert to CNF

P1

Given:

$\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

Using:

$$P \rightarrow Q \equiv \neg P \vee Q$$

we obtain:

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

Therefore:

C₁: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

P2

Given:

$\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

Eliminate implication:

$\neg(\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$

Using De Morgan's law:

$\neg\text{IrrigationNeeded} \vee \neg\text{CropWheat} \vee \text{ApplyDripMethod}$

Therefore:

$C_2: \neg\text{IrrigationNeeded} \vee \neg\text{CropWheat} \vee \text{ApplyDripMethod}$

P3

Given:

$\neg\text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Eliminate implication:

$\neg(\neg\text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$

Apply De Morgan's law:

$\text{ApplyDripMethod} \vee \neg\text{CropWheat} \vee \text{CropAtRisk}$

Therefore:

$C_3: \text{ApplyDripMethod} \vee \neg\text{CropWheat} \vee \text{CropAtRisk}$

Facts

$\text{SoilDry} = \text{True}$

$\text{CropWheat} = \text{True}$

These are already individual CNF clauses:

$C_4: \text{SoilDry}$

$C_5: \text{CropWheat}$

Final CNF Knowledge Base

$C_1: \neg\text{SoilDry} \vee \text{IrrigationNeeded}$

$C_2: \neg\text{IrrigationNeeded} \vee \neg\text{CropWheat} \vee \text{ApplyDripMethod}$

$C_3: \text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

$C_4: \text{SoilDry}$

$C_5: \text{CropWheat}$

(b) Resolution Refutation — Prove ApplyDripMethod

The theorem is:

ApplyDripMethod

For resolution refutation, we negate the goal.

Therefore:

$\neg \text{ApplyDripMethod}$

is added to the KB.

Step 1

From:

SoilDry

and:

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

resolve:

IrrigationNeeded

Step 2

We have:

IrrigationNeeded

and:

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

Resolve with IrrigationNeeded :

$\neg \text{CropWheat} \vee \text{ApplyDripMethod}$

Step 3

Resolve with:

CropWheat

Therefore:

ApplyDripMethod

Step 4: Contradiction

But for refutation, we added:

$\neg$ ApplyDripMethod

Now we have:

ApplyDripMethod

and:

$\neg$ ApplyDripMethod

Resolving them gives:

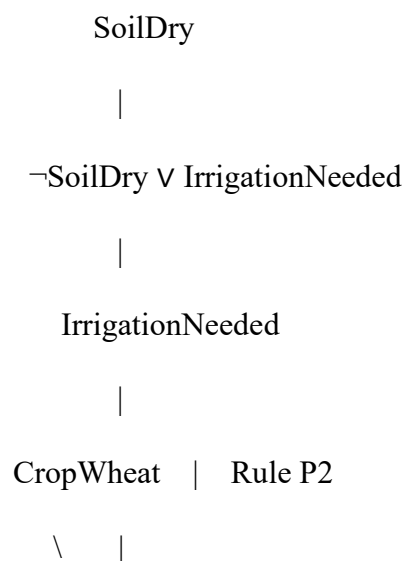
□

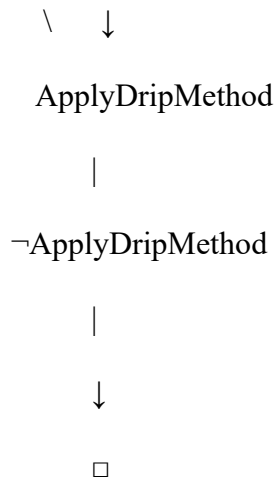
The empty clause means contradiction.

Therefore the original theorem is proved:

ApplyDripMethod

Resolution Proof Tree





Therefore, ApplyDripMethod is logically entailed by the KB.

(c) Remove SoilDry from the KB

Originally:

SoilDry

allowed us to derive:

IrrigationNeeded

Then:

$\text{IrrigationNeeded} \wedge \text{CropWheat}$

allowed us to derive:

ApplyDripMethod

If SoilDry is removed, Rule P1 cannot be triggered.

Therefore:

IrrigationNeeded

cannot be derived from the remaining facts.

Consequently, P2 cannot derive:

ApplyDripMethod

What about CropAtRisk?

Rule P3 is:

$\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Although:

$\text{CropWheat} = \text{True}$

is still available, we do not have a fact that says:

$\neg \text{ApplyDripMethod}$

The absence of a proof for ApplyDripMethod does not mean that its negation is true.

Therefore:

CropAtRisk

does not follow from the KB.

Final conclusion

$\text{SoilDry} \rightarrow \text{Not available}$

$\text{IrrigationNeeded} \rightarrow \text{Cannot be derived}$

$\text{ApplyDripMethod} \rightarrow \text{Cannot be derived}$

$\neg \text{ApplyDripMethod} \rightarrow \text{Not given}$

$\text{CropAtRisk} \rightarrow \text{Cannot be derived}$

This demonstrates the difference between failure to prove a statement and proving its negation.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1. $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell}[1,2]$
2. $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell}[1,1]$
3. $\text{Glitter}[x,y] \rightarrow \text{Gold is at}[x,y]$
4. $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

The Wumpus World case gives three percepts and four propositional rules.

Percepts:

Stench at [1,2]

Breeze at [1,1]

Glitter at [2,2]

(a) Belief State and Modus Ponens

Step 1: Initial Percepts

Stench[1,2]

Breeze[1,1]

Glitter[2,2]

Step 2: Apply Rule 1

Rule:

$\text{Stench}[1,2] \rightarrow \text{WumpusAdjacent}[1,2]$

Since Stench[1,2] is true, Modus Ponens gives:

WumpusAdjacent[1,2]

Step 3: Apply Rule 2

Rule:

$\text{Breeze}[1,1] \rightarrow \text{PitAdjacent}[1,1]$

Since $\text{Breeze}[1,1]$ is true:

$\text{PitAdjacent}[1,1]$

is derived.

Step 4: Apply Rule 3

Rule:

$\text{Glitter}[x,y] \rightarrow \text{Gold}[x,y]$

Given:

$\text{Glitter}[2,2]$

therefore:

$\text{Gold}[2,2]$

is derived.

Step 5: Rule 4

Rule:

$\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

However, the current percepts do not provide explicit negative facts such as:

$\neg \text{Wumpus}[x,y]$

and:

$\neg \text{Pit}[x,y]$

Therefore, Rule 4 cannot yet be fired for a specific cell.

Belief State

Percept / Conclusion	Status
Stench[1,2]	Known
Breeze[1,1]	Known
Glitter[2,2]	Known
Wumpus adjacent to [1,2]	Derived
Pit adjacent to [1,1]	Derived
Gold at [2,2]	Derived
Safe cells	Cannot yet be established

(b) Forward Chaining — Possible Wumpus and Pit Locations

The question asks for possible locations using the information available.

From Stench at [1,2]

The adjacent cells are:

[1,1]

[1,3]

[2,2]

Therefore, according to the given simplified rule, the Wumpus may be adjacent to [1,2], giving possible locations:

[1,1], [1,3], [2,2]

From Breeze at [1,1]

Adjacent cells are:

[1,2]

[2,1]

Therefore, the pit may be located at:

[1,2], [2,1]

From Glitter at [2,2]

The rule directly gives:

Glitter[2,2] $\rightarrow$ Gold[2,2]

Therefore:

Gold[2,2]

Reasoning Summary

Stench[1,2]

↓

Wumpus adjacent to [1,2]

↓

Possible Wumpus:

[1,1], [1,3], [2,2]

Breeze[1,1]

↓

Pit adjacent to [1,1]

↓

Possible Pit:

[1,2], [2,1]

Glitter[2,2]

↓

Gold[2,2]

(c) Is the Path [1,1] → [1,2] → [2,2] Safe?

The proposed path is:

[1,1] → [1,2] → [2,2]

We need to check each cell using the knowledge derived above.

Cell [1,1]

We know:

Breeze[1,1]

Therefore, according to Rule 2:

PitAdjacent[1,1]

So the agent has evidence that a pit may be adjacent to [1,1].

Therefore, safety cannot be established.

Cell [1,2]

We know:

Stench[1,2]

Therefore:

WumpusAdjacent[1,2]

So the agent knows that a Wumpus may be adjacent to this cell.

Again, the cell cannot be declared definitely safe.

Cell [2,2]

We know:

Glitter[2,2]

Therefore:

Gold[2,2]

The goal is located here, but the available KB does not establish:

$\neg$ Wumpus[2,2]

and:

$\neg$ Pit[2,2]

Therefore Rule 4 cannot establish:

Safe[2,2]

Final Decision

The path cannot be proven safe from the given knowledge base.

The important point is that the agent should not assume a cell is safe merely because no danger has been directly observed there.

For Rule 4 to establish safety, the KB must contain both:

$\neg \text{Wumpus}[x,y]$

and:

$\neg \text{Pit}[x,y]$

Neither is sufficiently established for all cells in the proposed path.

Therefore, a rational Wumpus World agent should not blindly follow the path. It should gather additional information or choose a path whose safety can be logically established.

FINAL SUMMARY

Case Study	Main AI Concept	Final Result
1. Medical Diagnosis	Propositional Logic, Forward & Backward Chaining	Flu and Pneumonia derived; Pneumonia goal proved
2. Traffic Management	FOL, Unification, Resolution	Proceed(CarA) proved
3. Agriculture	CNF, Resolution Refutation	ApplyDripMethod proved; after removing SoilDry, CropAtRisk does not follow
4. Wumpus World	Modus Ponens, Forward Chaining	Gold at [2,2]; possible danger exists; proposed path cannot be proven safe

The answers above directly follow the four case studies and sub-questions in the assessment. The intermediate inference steps show how conclusions are derived from facts and rules rather than being assumed.